

EXPERIMENT 5 — NETWORK MAPPING AND ANALYSIS USING NMAP

Aim

To familiarize students with the working of Nmap and perform network mapping, host discovery, port scanning, and service identification.

Description

Nmap (Network Mapper) is a network exploration and security-auditing tool. It can be used to obtain information about active hosts, IP addresses, open ports, closed ports, filtered ports, running services, service versions, operating systems, and network configuration. Nmap is commonly used by network administrators and security professionals for authorized network assessment.

Requirements

Platform	Requirements
Windows	Windows computer; Nmap; Command Prompt/PowerShell; authorized network
Kali Linux	Kali Linux; Nmap; Terminal; authorized network

PROCEDURE

A. WINDOWS PROCEDURE

Step 1 — Install Nmap

Install Nmap for Windows on the system.

Step 2 — Open Command Prompt or PowerShell

Open Command Prompt or Windows PowerShell to enter Nmap commands.

Step 3 — Verify installation

```
nmap --version
```

Step 4 — Scan an IP address

```
nmap 192.168.1.1
```

Step 5 — Scan an IP range

```
nmap 192.168.1.1-20
```

Step 6 — Scan a subnet

```
nmap 192.168.1.0/24
```

Step 7 — Scan selected ports

```
nmap -p 22,80,443 192.168.1.1
```

Step 8 — Detect services and versions

```
nmap -sV 192.168.1.1
```

Step 9 — Perform OS detection

```
nmap -O 192.168.1.1
```

Step 10 — Perform a detailed scan

```
nmap -A 192.168.1.1
```

B. KALI LINUX PROCEDURE

Step 1 — Install Nmap

```
sudo apt update  
sudo apt install nmap
```

Step 2 — Verify installation

```
nmap --version
```

Step 3 — Scan a host

```
nmap 192.168.1.1
```

Step 4 — Scan a subnet

```
nmap 192.168.1.0/24
```

Step 5 — Scan an IP range

```
nmap 192.168.1.1-20
```

Step 6 — Scan selected ports

```
nmap -p 22,80,443 192.168.1.1
```

Step 7 — Perform service detection

```
nmap -sV 192.168.1.1
```

Step 8 — Perform OS detection

```
sudo nmap -O 192.168.1.1
```

Step 9 — Perform a detailed scan

```
nmap -A 192.168.1.1
```

Important Nmap Options

Option	Function
-p	Specify ports
-sV	Service/version detection
-O	OS detection
-A	More comprehensive/aggressive detection
-Pn	Treat host as online and skip host discovery
-sT	TCP connect scan

RESULT ANALYSIS

Nmap successfully performs network mapping and port scanning. The scan results can show different port states.

Open

An open port indicates that an application is listening on the port and accepting connections.

Closed

A closed port indicates that the host is reachable, but no application is currently listening on that port.

Filtered

A filtered port indicates that a firewall or another filtering mechanism prevents Nmap from clearly determining the port state.

Service Detection

The -sV option can provide information about applications associated with open ports, including service/version information.

Operating System Detection

The -O option attempts to identify the operating system based on network responses from the target.

RESULT

Network mapping and port scanning were successfully performed using Nmap, and information regarding hosts, ports, and services was obtained.

OBSERVATIONS

1. Nmap can scan individual IP addresses.
2. Hostnames can also be scanned.
3. IP address ranges can be scanned.
4. Complete subnets can be examined.
5. Specific ports can be selected for scanning.
6. Open, closed, and filtered ports can be observed.
7. Service versions can be identified using service detection.
8. OS detection can provide additional information.
9. Nmap works on both Windows and Kali Linux.
10. Windows uses Command Prompt or PowerShell for Nmap commands.
11. Kali Linux uses the Linux terminal.
12. Nmap provides different scanning options depending on the required information.
13. -p can be used to specify particular ports.
14. -sV can be used for service and version detection.
15. -O can be used for operating-system detection.
16. -A provides a more comprehensive scan.
17. -Pn can treat a host as online and skip host discovery.
18. -sT performs a TCP connect scan.
19. Port states provide information about the accessibility of services.
20. Nmap should only be used against systems and networks for which proper authorization has been obtained.

Practical Flow: Install Nmap → Select authorized target → Perform host/port scan → Use service or OS detection if required → Analyse the output → Record the discovered information.